

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks: 50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

(192511488)

I Varsha Vardhini. S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Varsha Vardhini

78%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

MARKS ALLOCATION :

	Problem Understanding (20%)	Method Selection (20%)	Solution process (25%)	Accuracy (20%)	Interpretation of result (15%)
9.5 Q1	2	1.5	2.5	2	1.5
8.5 Q2	1.5	2	2	2	1
7 Q3	1	1	2.5	1	1.5
7 Q4	1.5	1.5	2	1	1
7 Q5	2	1	1	1.5	1.5
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1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

ANSWER :

Given

- Original Network: 192.168.20.0/24
- Lab A: /26
- Lab B: /27
- Lab C: /28

Formulae Used

Host bits = 32 – Prefix length

Total IP addresses = $2^{(\text{Host bits})}$

Usable host addresses = Total IP addresses – 2

Block size = 256 – Last octet of subnet mask

Lab A – /26

Subnet mask = 255.255.255.192

Host bits : $32 - 26 = 6$ bits

Total addresses : $2^6 = 64$ addresses

Usable host addresses : $64 - 2 = 62$ hosts

Block size : $256 - 192 = 64$

Therefore, the first subnet is : 192.168.20.0 – 192.168.20.63

So:

- Network Address = 192.168.20.0
- Valid Host Range = 192.168.20.1 – 192.168.20.62
- Broadcast Address = 192.168.20.63
- Usable Hosts = 62

Lab B – /27

Lab A occupies 192.168.20.0 – 192.168.20.63.

Therefore, Lab B starts at 192.168.20.64.

Subnet mask = 255.255.255.224

Host bits : $32 - 27 = 5$ bits

Total addresses : $2^5 = 32$ addresses

Usable host addresses : $32 - 2 = 30$ hosts

Block size : $256 - 224 = 32$

Therefore, the subnet is : 192.168.20.64 – 192.168.20.95

So:

- Network Address = 192.168.20.64
- Valid Host Range = 192.168.20.65 – 192.168.20.94
- Broadcast Address = 192.168.20.95
- Usable Hosts = 30

Lab C – /28

Lab B occupies 192.168.20.64 – 192.168.20.95.

Therefore, Lab C starts at 192.168.20.96.

Subnet mask = 255.255.255.240

Host bits : $32 - 28 = 4$ bits

Total addresses : $2^4 = 16$ addresses

Usable host addresses : $16 - 2 = 14$ hosts

Block size : $256 - 240 = 16$

Therefore, the subnet is : 192.168.20.96 – 192.168.20.111

So:

- Network Address = 192.168.20.96
- Valid Host Range = 192.168.20.97 – 192.168.20.110
- Broadcast Address = 192.168.20.111
- Usable Hosts = 14

FINAL ANSWER :

Lab	Subnet	Network Address	Valid Host IP Range	Broadcast Address	Usable Host Addresses
Lab A	/26	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	62
Lab B	/27	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	30
Lab C	/28	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	14

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

ANSWER :

Given

- System 1 IP: 192.168.1.130
- System 2 IP: 192.168.1.190
- Subnet Mask: 255.255.255.192 (/26)

Formulae Used

$$\text{Host bits} = 32 - \text{Prefix length}$$

$$\text{Total addresses} = 2^{(\text{Host bits})}$$

$$\text{Usable hosts} = \text{Total addresses} - 2$$

$$\text{Block size} = 256 - \text{Last octet of subnet mask}$$

Step 1: Find the /26 Subnet Blocks

$$\text{Subnet mask} = 255.255.255.192$$

$$\text{Block size} : 256 - 192 = 64$$

Therefore, the subnet blocks in the last octet are:

- 0 – 63
- 64 – 127
- 128 – 191
- 192 – 255

Both IP addresses, 130 and 190, fall within the 128 – 191 block.

Therefore, both systems belong to the same subnet.

Step 2: System 1 – 192.168.1.130

The IP 192.168.1.130 falls within : 192.168.1.128 – 192.168.1.191

Therefore:

- Network Address: 192.168.1.128
- Valid Host Range: 192.168.1.129 – 192.168.1.190
- Broadcast Address: 192.168.1.191

Check

192.168.1.130 lies between 192.168.1.129 and 192.168.1.190.

Therefore : System 1 IP is VALID

Step 3: System 2 – 192.168.1.190

The IP 192.168.1.190 also falls within : 192.168.1.128 – 192.168.1.191

Therefore:

- Network Address: 192.168.1.128
- Valid Host Range: 192.168.1.129 – 192.168.1.190
- Broadcast Address: 192.168.1.191

Check

192.168.1.190 is the last valid host address before the broadcast address.

Therefore : System 2 IP is VALID

Step 4: Number of Usable Hosts

Host bits : $32 - 26 = 6$ bits

Total addresses : $2^6 = 64$

Usable hosts : $64 - 2 = 62$

Therefore, the subnet has : 62 usable host addresses

System	IP Address	Subnet Range	Network Address	Valid Host Range	Broadcast Address	IP Valid?
System 1	192.168.1.130	192.168.1.128 – 192.168.1.191	192.168.1.128	192.168.1.129 – 192.168.1.190	192.168.1.191	Yes
System 2	192.168.1.190	192.168.1.128 – 192.168.1.191	192.168.1.128	192.168.1.129 – 192.168.1.190	192.168.1.191	Yes

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

ANSWER :

Given

- Original Network: 172.16.0.0/16
- HR → /25
- Sales → /26
- IT → /27
- Admin → /28

Formulae Used

Host bits = 32 – Prefix length

Total addresses = $2^{(\text{Host bits})}$

Usable hosts = Total addresses – 2

Block size = 256 – Last octet of subnet mask

HR – /25

Subnet mask = 255.255.255.128

Host bits : $32 - 25 = 7$

Total addresses : $2^7 = 128$

Usable hosts : $128 - 2 = 126$

Block size : $256 - 128 = 128$

Therefore : 172.16.0.0 – 172.16.0.127

- Network Address: 172.16.0.0
- Valid Host Range: 172.16.0.1 – 172.16.0.126
- Broadcast Address: 172.16.0.127
- Usable Hosts: 126

Sales – /26

HR occupies 172.16.0.0 – 172.16.0.127.

Therefore, Sales starts at 172.16.0.128.

Subnet mask = 255.255.255.192

Host bits : $32 - 26 = 6$

Total addresses : $2^6 = 64$

Usable hosts : $64 - 2 = 62$

Block size : $256 - 192 = 64$

Therefore : 172.16.0.128 – 172.16.0.191

- Network Address: 172.16.0.128
- Valid Host Range: 172.16.0.129 – 172.16.0.190
- Broadcast Address: 172.16.0.191
- Usable Hosts: 62

IT – /27

Sales occupies 172.16.0.128 – 172.16.0.191.

Therefore, IT starts at 172.16.0.192.

Subnet mask = 255.255.255.224

Host bits : $32 - 27 = 5$

Total addresses : $2^5 = 32$

Usable hosts : $32 - 2 = 30$

Block size : $256 - 224 = 32$

Therefore : 172.16.0.192 – 172.16.0.223

- Network Address: 172.16.0.192
- Valid Host Range: 172.16.0.193 – 172.16.0.222
- Broadcast Address: 172.16.0.223
- Usable Hosts: 30

Admin – /28

IT occupies 172.16.0.192 – 172.16.0.223.

Therefore, Admin starts at 172.16.0.224.

Subnet mask = 255.255.255.240

Host bits : $32 - 28 = 4$

Total addresses : $2^4 = 16$

Usable hosts : $16 - 2 = 14$

Block size : $256 - 240 = 16$

Therefore : 172.16.0.224 – 172.16.0.239

- Network Address: 172.16.0.224
- Valid Host Range: 172.16.0.225 – 172.16.0.238
- Broadcast Address: 172.16.0.239
- Usable Hosts: 14

Remaining Unused IP Range

The original /16 network is : 172.16.0.0 – 172.16.255.255

Admin ends at : 172.16.0.239

Therefore, the remaining unused range is : 172.16.0.240 – 172.16.255.255

Department	Subnet	Network Address	Valid Host IP Range	Broadcast Address	Usable Host Addresses
HR	/25	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales	/26	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	62
IT	/27	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin	/28	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	14

Original Network	Allocated Up To	Remaining Unused Range
172.16.0.0/16	172.16.0.239	172.16.0.240 – 172.16.255.255

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

ANSWER :

Given

- Original network = 172.16.10.0/24
- New subnet mask = /27
- IP 1 = 172.16.10.78
- IP 2 = 172.16.10.95

Formulae Used

Borrowed bits = New prefix – Original prefix

Number of subnets = $2^{(\text{Borrowed bits})}$

Host bits = 32 – New prefix

Usable hosts per subnet = $2^{(\text{Host bits})} - 2$

Block size = 256 – Last octet of subnet mask

Number of Subnets

Original prefix = /24

New prefix = /27

Borrowed bits : $27 - 24 = 3$ bits

Number of subnets : $2^3 = 8$ subnets

Therefore, 8 subnets are created.

Usable Hosts per Subnet

Host bits : $32 - 27 = 5$ bits

Total addresses per subnet : $2^5 = 32$ addresses

Usable hosts : $32 - 2 = 30$ usable hosts

Therefore, each /27 subnet supports 30 usable hosts.

Find the Subnet of 172.16.10.78

Subnet mask /27 = 255.255.255.224

Block size : $256 - 224 = 32$

Therefore, the subnet ranges are:

172.16.10.0 – 172.16.10.31

172.16.10.32 – 172.16.10.63

172.16.10.64 – 172.16.10.95

172.16.10.96 – 172.16.10.127

...

The IP 172.16.10.78 falls within: 172.16.10.64 – 172.16.10.95

Therefore:

- Network Address: 172.16.10.64
- Valid Host Range: 172.16.10.65 – 172.16.10.94
- Broadcast Address: 172.16.10.95

Check 172.16.10.95

The broadcast address of this subnet is : 172.16.10.95

A broadcast address cannot be assigned to a host.

Therefore : 172.16.10.95 belongs to the subnet range, but it is NOT a valid host IP address.

So the answer is: Yes, it belongs to the same subnet, but it is the broadcast address.

Parameter	Answer
Original Network	172.16.10.0/24
New Subnet Mask	/27 (255.255.255.224)
Borrowed Bits	3
Total Number of Subnets	8
Total Addresses per Subnet	32
Usable Hosts per Subnet	30
Subnet containing 172.16.10.78	172.16.10.64/27
Network Address	172.16.10.64
Valid Host Range	172.16.10.65 – 172.16.10.94
Broadcast Address	172.16.10.95

Does 172.16.10.95 belong to the subnet? Yes

Is 172.16.10.95 a valid host address? No – it is the broadcast address

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

ANSWER :

Given

IPv6 Address = 2001:0db8:0000:0000:0000:ff00:0042:8329

Prefix Length = /64

Compress the IPv6 Address

IPv6 compression rules:

- Remove leading zeros from each hexadecimal group.
- Replace the longest consecutive sequence of 0000 groups with ::.

Given : 2001:0db8:0000:0000:0000:ff00:0042:8329

Remove leading zeros : 2001:db8:0:0:0:ff00:42:8329

The three consecutive 0 groups can be replaced by ::.

Compressed Address : 2001:db8::ff00:42:8329

Expand the Compressed Address

Compressed : 2001:db8::ff00:42:8329

An IPv6 address contains 8 groups.

Currently, we have : 2001 | db8 | :: | ff00 | 42 | 8329

There are 5 groups written, so :: represents 3 groups of 0000.

Therefore : 2001:0db8:0000:0000:0000:ff00:0042:8329

Expanded Address : 2001:0db8:0000:0000:0000:ff00:0042:8329

Network Prefix

The prefix length is /64.

IPv6 has 128 bits in total.

Therefore, the first 64 bits represent the network portion.

The first four groups are : 2001 : 0db8 : 0000 : 0000

Each group = 16 bits.

$$4 \times 16 = 64 \text{ bits}$$

Network Prefix : 2001:0db8:0000:0000/64

Or in compressed form : 2001:db8::/64

Total Possible Host Addresses

Find host bits : Host bits = $128 - 64 = 64$ bits

Step 2: Calculate possible host addresses

Formula : Total host addresses = $2^{(\text{Host bits})} = 2^{64}$
= 18,446,744,073,709,551,616

Therefore, the /64 network supports:

18,446,744,073,709,551,616 possible host addresses

For IPv6, we do not subtract 2 as we commonly do in IPv4 subnet calculations.

FINAL ANSWER :

Parameter	Answer
Given IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed Address	2001:db8::ff00:42:8329
Expanded Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix Length	/64
Network Prefix	2001:0db8:0000:0000/64
Host Bits	$128 - 64 = 64$ bits
Total Possible Host Addresses	$2^{64} = 18,446,744,073,709,551,616$

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